ECDLP and special attack types

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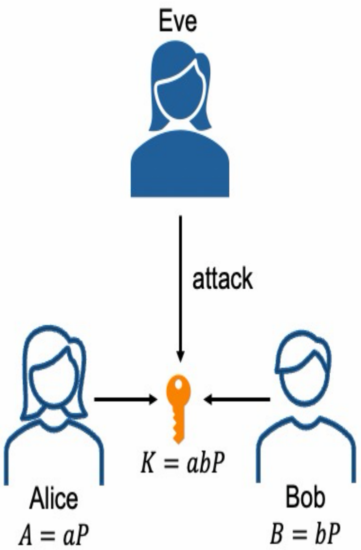
Background

- ECDLP is a hard problem where we try to find the private key d in $Q = dP$ where $P, Q \in E(GF(p))$ and p is a prime.
- An algorithm's efficiency depends on two main factors: the amount of memory usage and the number of steps. We calculate these values with complexity theory.
- Weil pairing is a function that transforms two points from the elliptic curve into a number: $e_n(P, Q) = \zeta$

Goal

- What are the attacks against the ECDLP?
- What are their algorithms and how efficient are they?
- In what ways are they powerful?

Eve attacking $K = abP$



Baby-Step Giant-Step (BSGS) Attack

Baby-Step (BS)
➤ Store all the values of iP

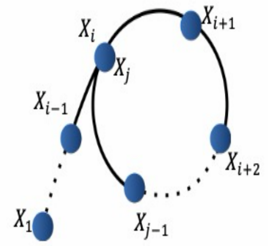
✚

Giant-Step (GS)
➤ Find j such that $Q - njP = iP$

We try to find the private key as

$$d = jn + i$$
$$\Rightarrow Q = dP = (jn + i)P = jnP + iP$$
$$\Rightarrow Q - njP = iP$$

Pollard's rho Attack



This algorithm calculates d with collusions:

$$X_i = a_iP + b_iQ = a_jP + b_jQ = X_j$$
$$a_iP + b_idP = a_jP + b_jdP$$
$$d \equiv \frac{a_i - a_j}{b_j - b_i} \pmod p$$

MOV Attack

$$\beta = e_n(Q, T) = e_n(dP, T) = e_n(Q, T)^d = \alpha^d$$

✚

➤ define a base

$\mathcal{B} = \{-1, 2, \dots, p_r\}$ where p is a prime

➤ find constants k and e^r

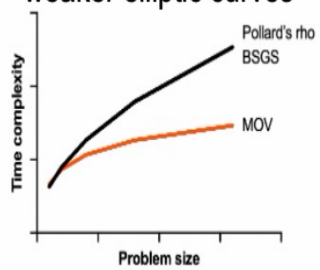
$\alpha^k \equiv (-1)^{e_0} 2^{e_1} \cdots {p_r}^{e_r} \pmod p$

and find $log_\alpha p^r$

➤ if $\alpha^s \beta \equiv (-1)^{m_0} 2^{m_1} \cdots {p_r}^{m_r} \pmod p$

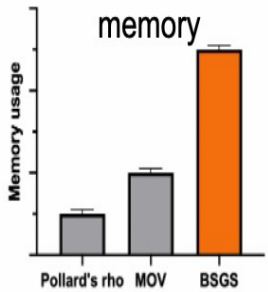
➤ then $d \equiv m_0 log_\alpha(-1) + \cdots m_r log_\alpha p^r - s$

MOV attack works faster on weaker elliptic curves



- On supersingular elliptic curves, where the embedding degree k is small, MOV attack requires fewer steps.

BSGS uses the most memory



- BSGS uses the most memory: $O(\sqrt{N})$
- MOV attack uses less than BSGS: $O(\exp[c'(\ln(p^k))^{\frac{1}{3}}(\ln\ln(p^k))^{\frac{2}{3}}])$ where $c' < c$
- Pollard's rho is the most memory-efficient: $O(1)$

Conclusion

- The MOV attack is the fastest attack type when working on supersingular elliptic curves.
- ECC is set on ordinary curves instead of supersingular.
- Pollard's rho attack is the most efficient of these three special attacks, as it requires less memory and fewer steps.